

Lecture 26/Module 3

Sinusoidal Steady State Analysis

11/09/2026

Dr. Sudipta Kumar Sarkar

Assistant Professor (Research), SENSE



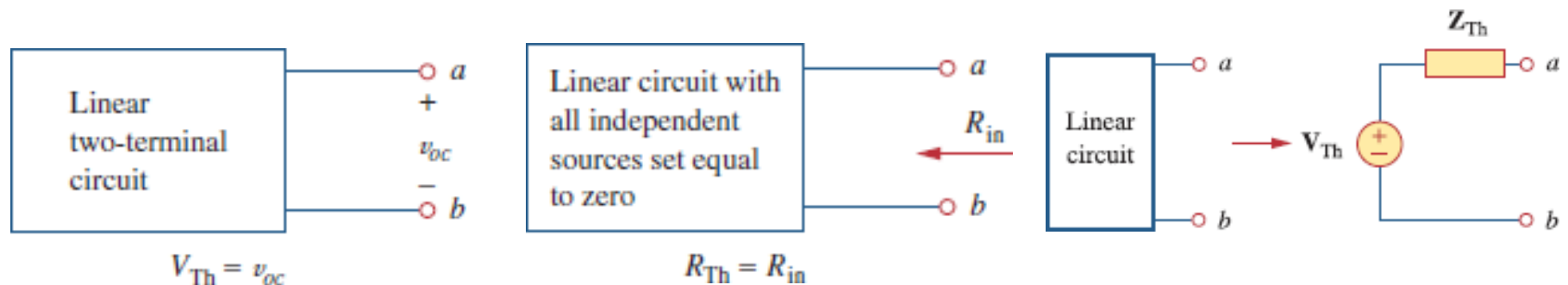
School of Electronics Engineering (SENSE)
VIT, Vellore, India

Today's Topics

- Phasor – Thevenin's Theorem
- Phasor – Norton's Theorem
- Phasor – Maximum Power Theorem
- Instantaneous Power, Average Power, Complex Power

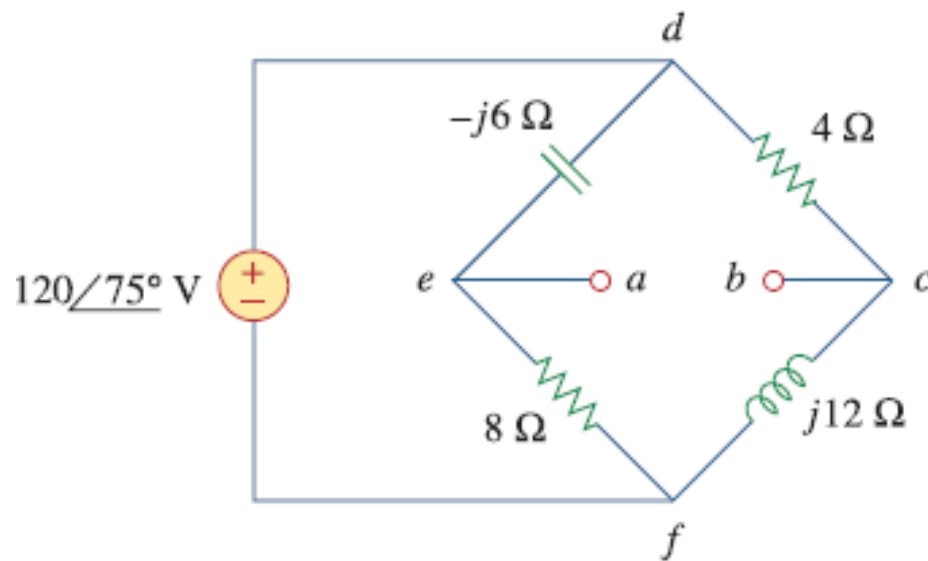
Thevenin's Equivalent Circuit

Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



Numerical -1

Obtain the Thevenin equivalent at terminals $a-b$ of the circuit



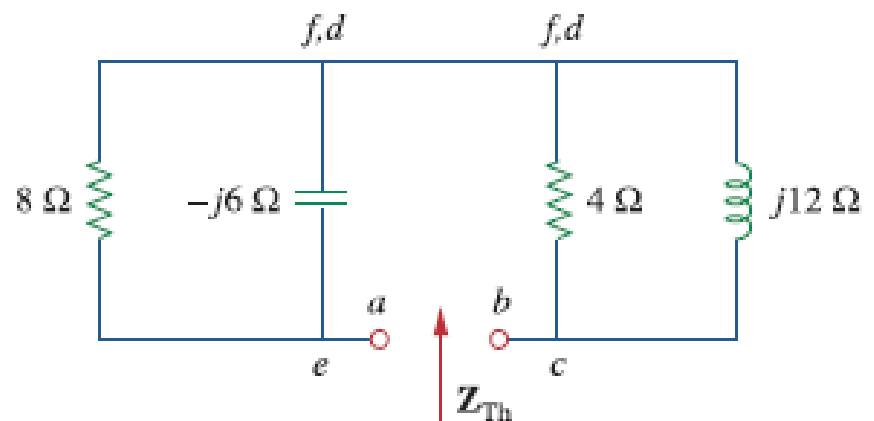
Solution:

To find R_{Th} :

$$Z_1 = -j6 \parallel 8 = \frac{-j6 \times 8}{8 - j6} = 2.88 - j3.84 \Omega$$

$$Z_2 = 4 \parallel j12 = \frac{j12 \times 4}{4 + j12} = 3.6 + j1.2 \Omega$$

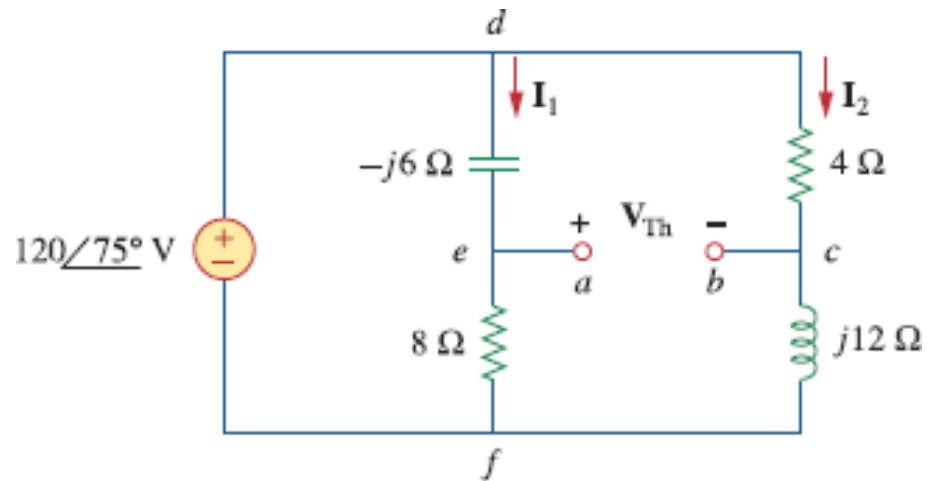
$$Z_{Th} = Z_1 + Z_2 = 6.48 - j2.64 \Omega$$



- To find V_{Th} :

$$I_1 = \frac{120 \angle 75^\circ}{8 - j6} \text{ A}$$

$$I_2 = \frac{120 \angle 75^\circ}{4 + j12} \text{ A}$$



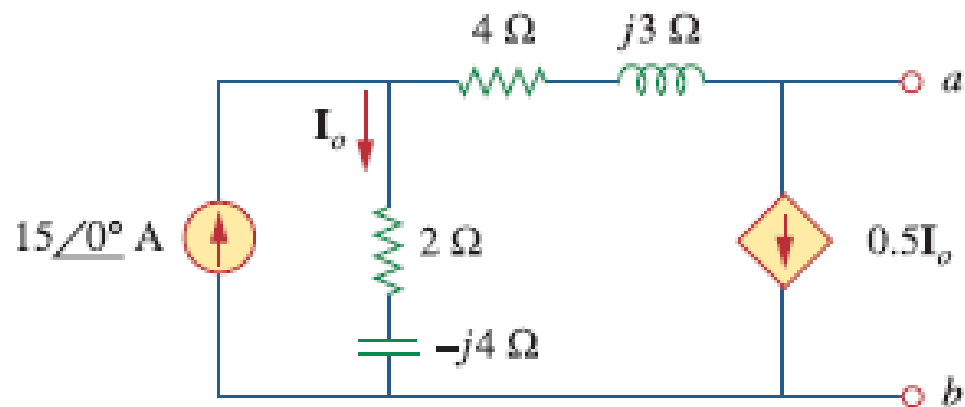
Applying KVL around loop *bcdeab*

$$\begin{aligned} V_{Th} = 4I_2 + j6I_1 &= \frac{480 \angle 75^\circ}{4 + j12} + \frac{720 \angle 75^\circ + 90^\circ}{8 - j6} \\ &= 37.95 \angle 3.43^\circ + 72 \angle 201.87^\circ \\ &= -28.936 - j24.55 = 37.95 \angle 220.31^\circ \text{ V} \end{aligned}$$

Note: Another method:
Use voltage division at a and b, then subtract the results to get V_{th} .

Numerical -2

Find the Thevenin equivalent of the circuit



Solution:

- To find V_{Th} :

we apply KCL at node 1

$$15 = I_o + 0.5I_o$$

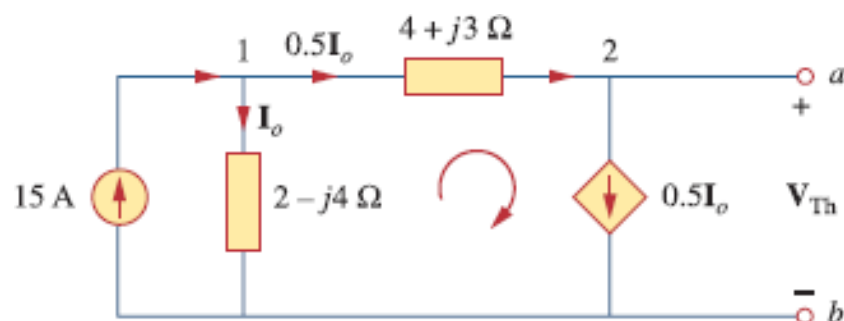
$$I_o = 10 \text{ A}$$

Applying KVL to the loop on the right-hand side

$$-I_o(2 - j4) + 0.5I_o(4 + j3) + V_{Th} = 0$$

$$V_{Th} = 10(2 - j4) - 5(4 + j3) = -j55$$

$$V_{Th} = 55 \angle -90^\circ \text{ V}$$



To find R_{Th} :

- Remove the independent current source
- Connect any arbitrary current source across the open end, find the voltage across it.
- Applying KCL at node V_s

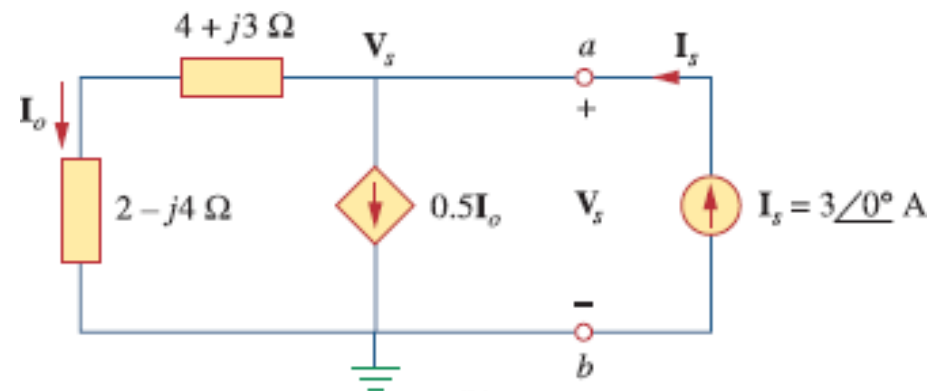
$$3 = I_o + 0.5I_o \Rightarrow I_o = 2 \text{ A}$$

Applying KVL to the outer loop

$$V_s = I_o(4 + j3 + 2 - j4) = 2(6 - j)$$

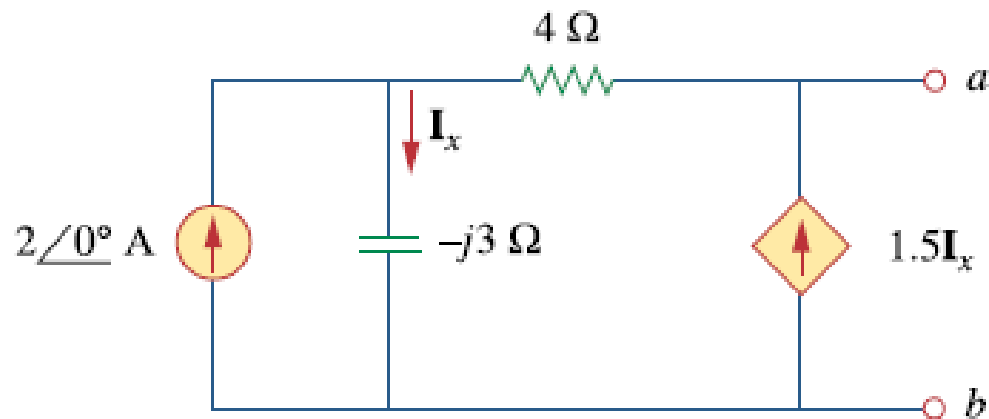
The Thevenin impedance is

$$Z_{Th} = \frac{V_s}{I_s} = \frac{2(6 - j)}{3} = 4 - j0.6667 \Omega$$

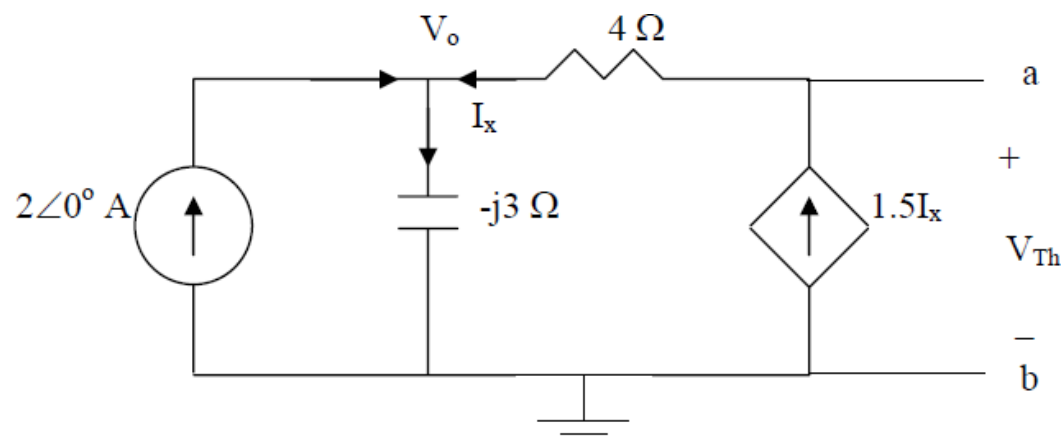


Numerical -3

Find the Thevenin equivalent at terminals a - b of the circuit



To find V_{Th} , consider the circuit below

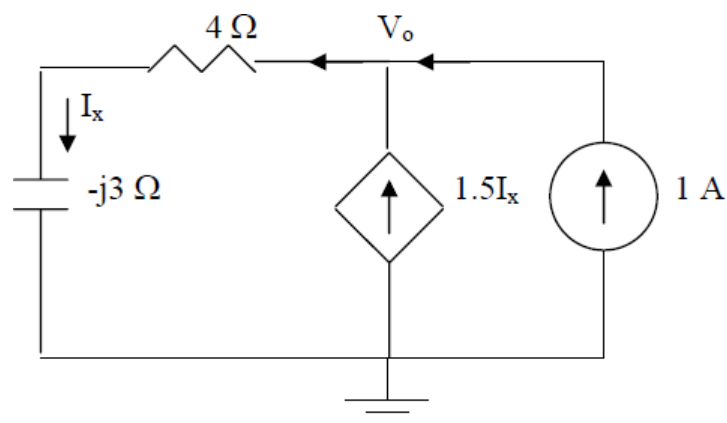


$$2 + 1.5I_x = I_x \longrightarrow I_x = -4$$

$$\text{But } V_o = -j3I_x = j12$$

$$V_{Th} = V_o + 6I_x = \underline{j12 - 24 \text{ V}}$$

To find Z_{Th} , consider the circuit shown below.



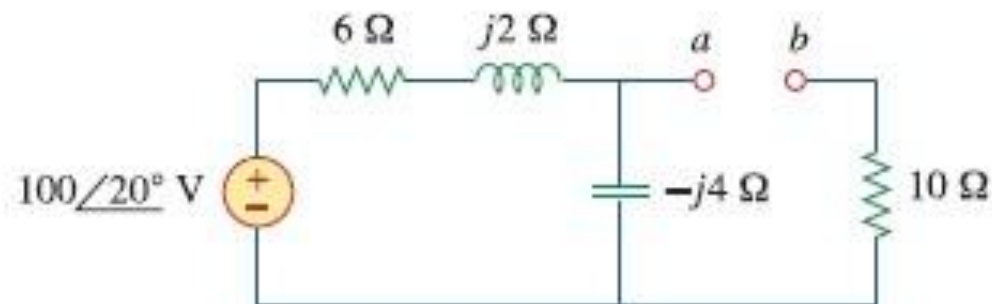
$$1 + 1.5 I_x = I_x \quad \Rightarrow \quad I_x = -2$$

$$-V_o + I_x(4 - j3) = 0 \quad \longrightarrow \quad V_o = -8 + j6$$

$$Z_{Th} = \frac{V_o}{1} = \underline{\underline{-8 + j6 \Omega}}$$

Homework problem -1

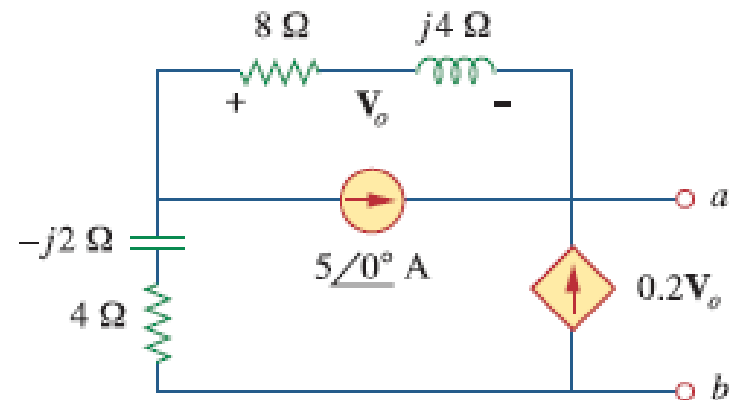
Find the Thevenin equivalent at terminals a - b of the circuit in Fig.



Answer: $Z_{Th} = 12.4 - j3.2\ \Omega$, $V_{Th} = 63.24\angle -51.57^\circ$ V.

Homework problem -2

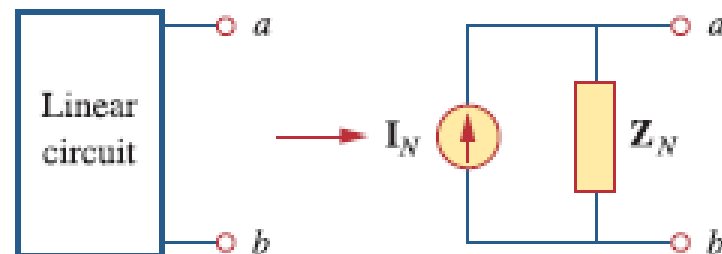
Determine the Thevenin equivalent of the circuit in Fig.



Answer: $Z_{Th} = 4.473\angle -7.64^\circ\ \Omega$, $V_{Th} = 7.35\angle 72.9^\circ\text{ V}$.

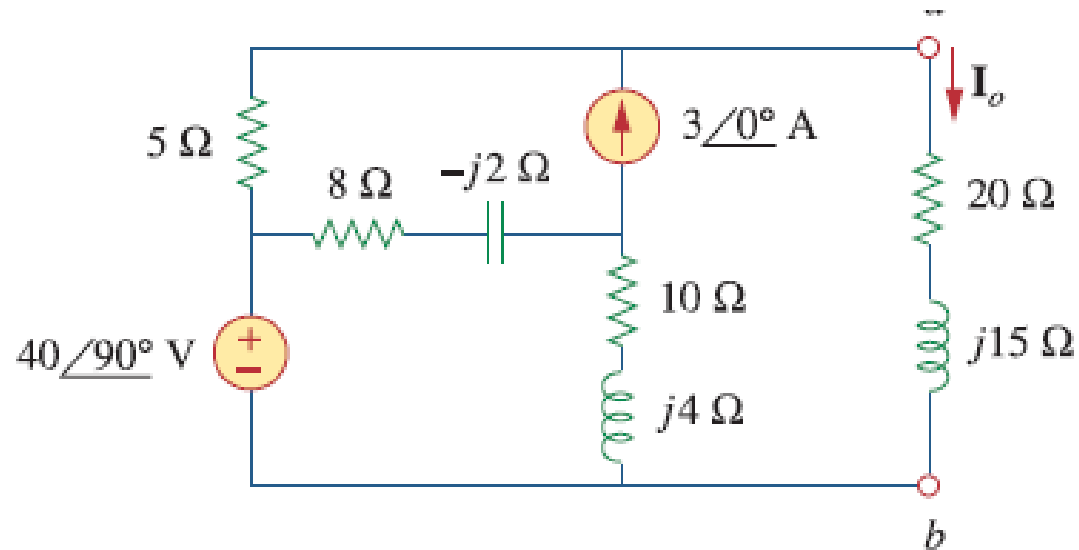
Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.



Numerical -1

Obtain current $\mathbf{I}_o$ in Fig. using Norton's theorem.



Solution:

To find Z_N :

$$Z_N = 5 \Omega$$

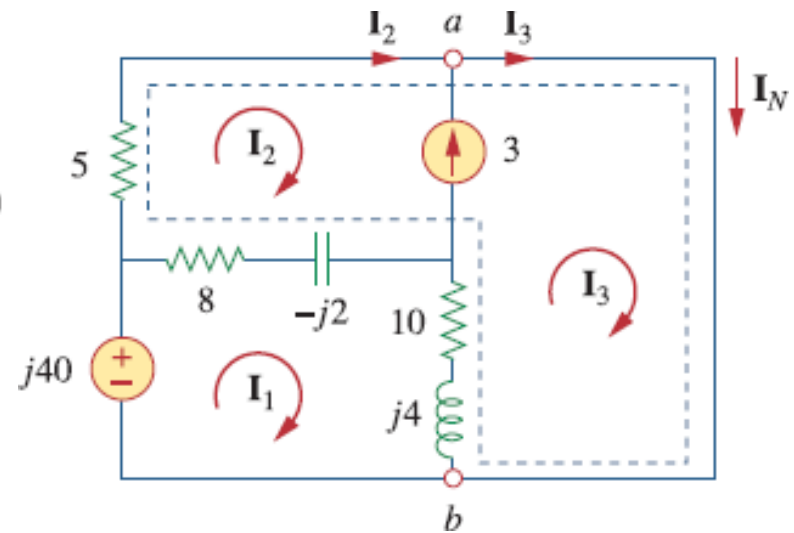
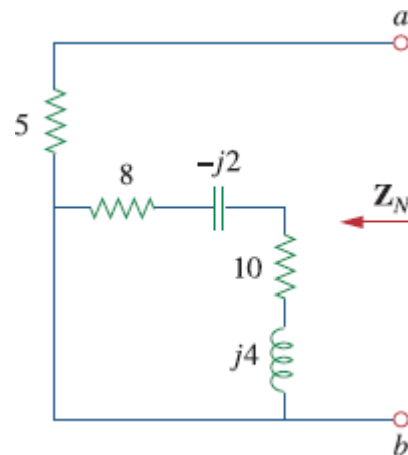
To find I_N :

For mesh 1,

$$-j40 + (18 + j2)\mathbf{I}_1 - (8 - j2)\mathbf{I}_2 - (10 + j4)\mathbf{I}_3 = 0$$

For the supermesh,

$$(13 - j2)\mathbf{I}_2 + (10 + j4)\mathbf{I}_3 - (18 + j2)\mathbf{I}_1 = 0$$



At node a ,

$$\mathbf{I}_3 = \mathbf{I}_2 + 3$$

Adding Mesh 1 and super mesh equations,

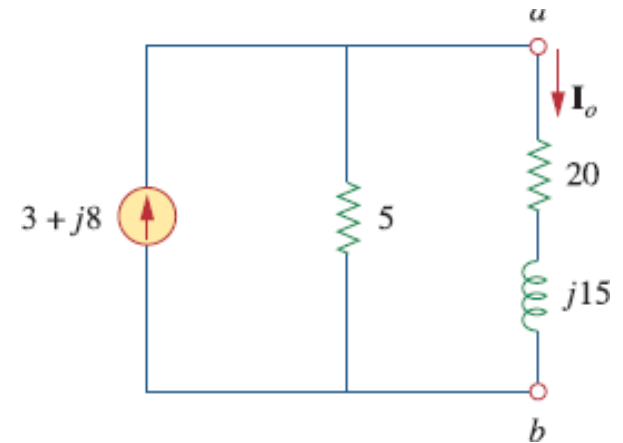
$$-j40 + 5\mathbf{I}_2 = 0 \quad \Rightarrow \quad \mathbf{I}_2 = j8$$

$$\mathbf{I}_3 = \mathbf{I}_2 + 3 = 3 + j8$$

The Norton current is

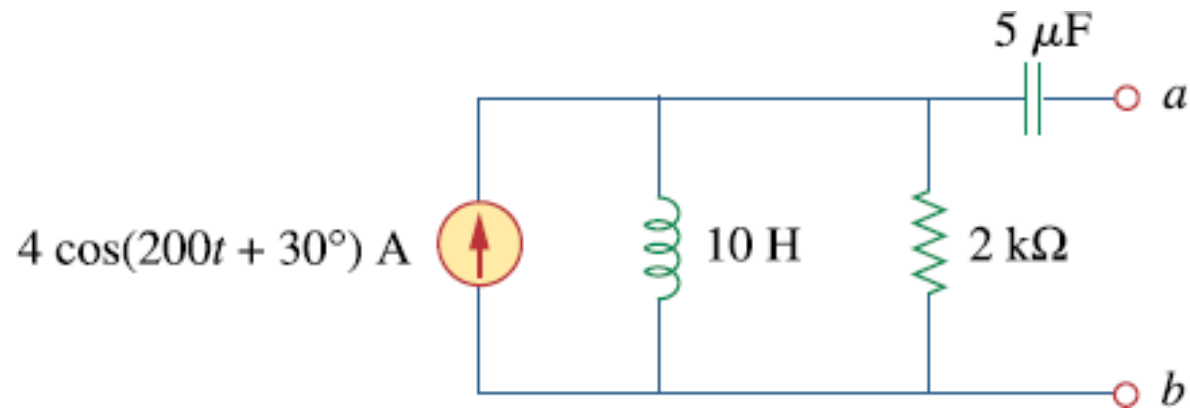
$$\mathbf{I}_N = \mathbf{I}_3 = (3 + j8) \text{ A}$$

$$\mathbf{I}_o = \frac{5}{5 + 20 + j15} \mathbf{I}_N = \frac{3 + j8}{5 + j3} = 1.465 \angle 38.48^\circ \text{ A}$$



Numerical -2

- Obtain the Norton's equivalent for the circuit.



- Solution:

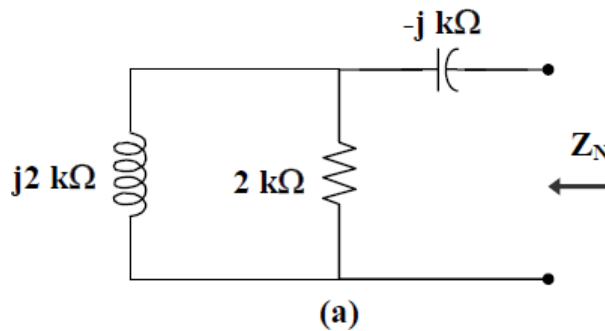
Transform the circuit to the frequency domain.

$$4 \cos(200t + 30^\circ) \longrightarrow 4 \angle 30^\circ, \quad \omega = 200$$

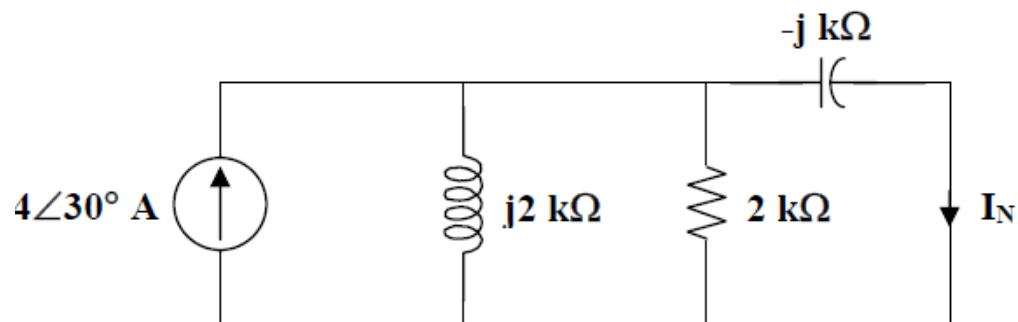
$$10 \text{ H} \longrightarrow j\omega L = j(200)(10) = j2 \text{ k}\Omega$$

$$5 \text{ }\mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(200)(5 \times 10^{-6})} = -j \text{ k}\Omega$$

Z_N is found using the circuit in Fig. (a).



$$Z_N = -j + 2 \parallel j2 = -j + 1 + j = 1 \text{ k}\Omega$$



$$j2 \parallel 2 = 1 + j$$

By the current division principle,

$$\mathbf{I}_N = \frac{1 + j}{1 + j - j} (4\angle 30^\circ) = 5.657\angle 75^\circ$$

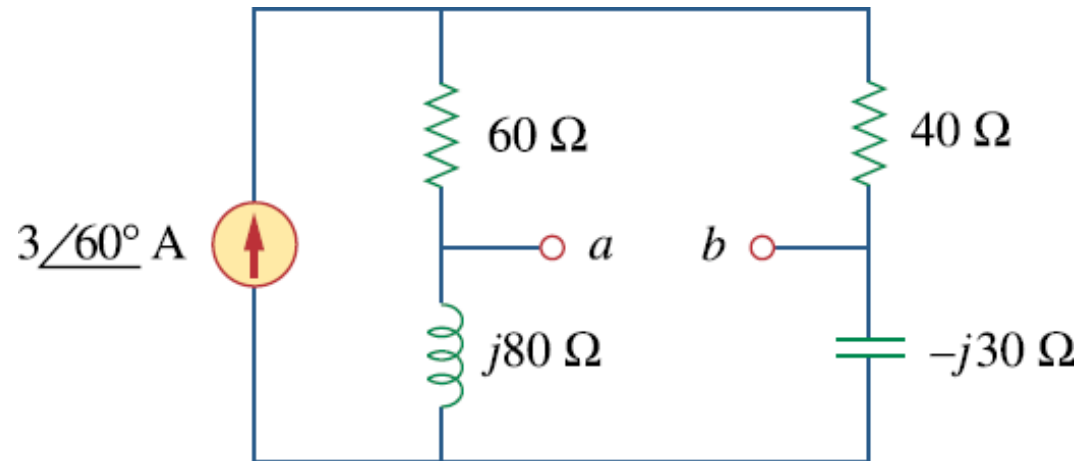
Therefore,

$$i_N = \underline{\underline{5.657 \cos(200t + 75^\circ) \text{ A}}}$$

$$\mathbf{Z}_N = \underline{\underline{1 \text{ k}\Omega}}$$

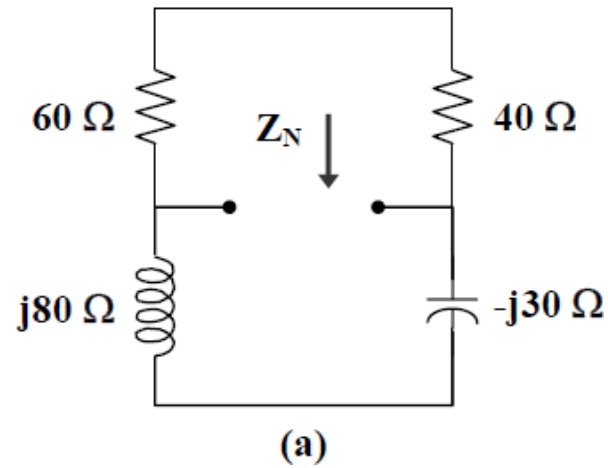
Numerical -3

- Obtain the Norton's equivalent for the circuit.



- Solution:

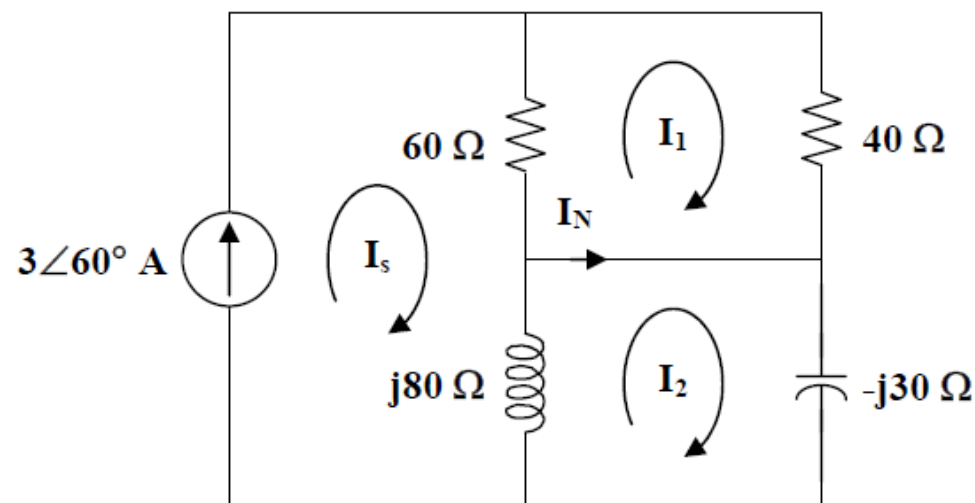
Z_N is obtained from the circuit in Fig. (a).



$$Z_N = (60 + 40) \parallel (j80 - j30) = 100 \parallel j50 = \frac{(100)(j50)}{100 + j50}$$

$$Z_N = 20 + j40 = \underline{\underline{44.72 \angle 63.43^\circ \Omega}}$$

To find $\mathbf{I}_N$, consider the circuit in Fig. (b).



(b)

$$\mathbf{I}_s = 3\angle 60^\circ$$

For mesh 1,

$$100\mathbf{I}_1 - 60\mathbf{I}_s = 0$$

$$\mathbf{I}_1 = 1.8\angle 60^\circ$$

For mesh 2,

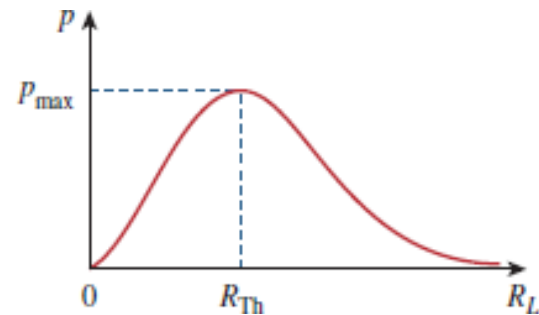
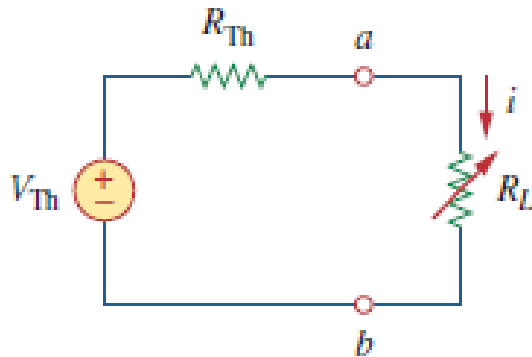
$$(j80 - j30)\mathbf{I}_2 - j80\mathbf{I}_s = 0$$

$$\mathbf{I}_2 = 4.8\angle 60^\circ$$

$$\mathbf{I}_N = \mathbf{I}_2 - \mathbf{I}_1 = \underline{\underline{3\angle 60^\circ \text{ A}}}$$

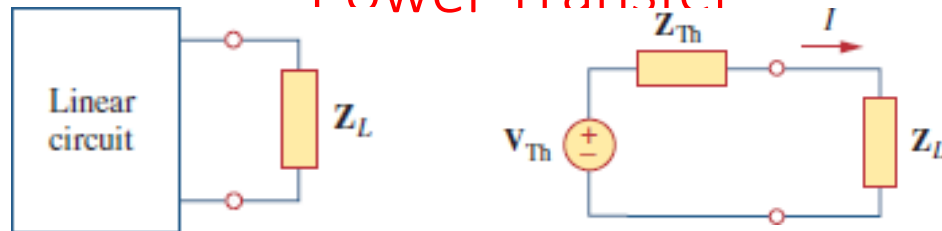
Maximum Power Transfer Theorem

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).



$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

Maximum Average Power Transfer



$$Z_{Th} = R_{Th} + jX_{Th}$$

$$Z_L = R_L + jX_L$$

$$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$$

For **maximum average power transfer**, the load impedance $\mathbf{Z}_L$ must be equal to the complex conjugate of the Thevenin impedance $\mathbf{Z}_{Th}$.

This result is known as the *maximum average power transfer theorem* for the sinusoidal steady state.

Setting $R_L = R_{Th}$ and $X_L = -X_{Th}$

$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}}$$

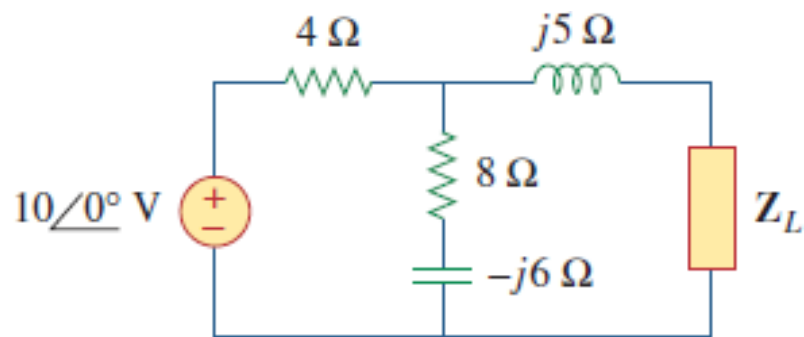
$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |Z_{Th}|$$

Numerical

- 1

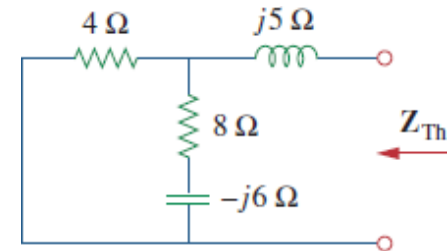
Determine the load impedance Z_L that maximizes the average power drawn from the circuit

What is the maximum average power?



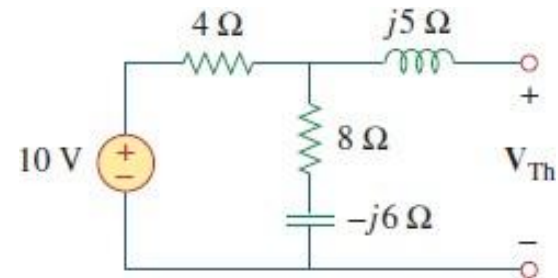
To find Z_{Th} :

$$Z_{Th} = j5 + 4 \parallel (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} \\ = 2.933 + j4.467 \Omega$$



To find V_{Th} :

$$V_{Th} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454 \angle -10.3^\circ \text{ V}$$



The load impedance draws the maximum power from the circuit when

$$Z_L = Z_{Th}^* = 2.933 - j4.467 \Omega$$

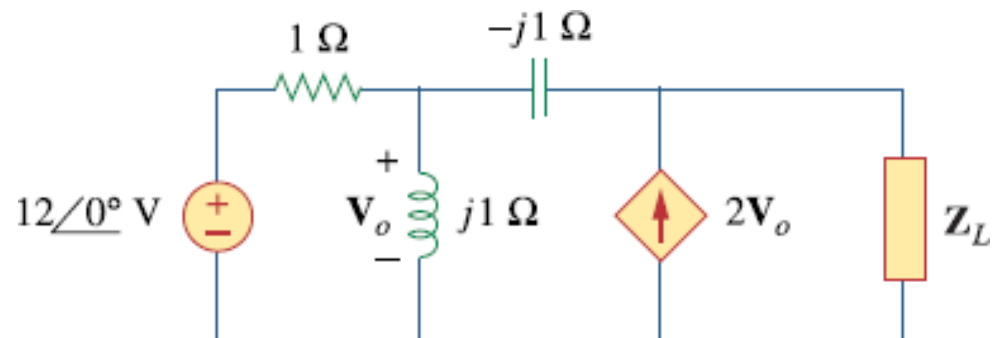
the maximum average power is

$$P_{\max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{(7.454)^2}{8(2.933)} = 2.368 \text{ W}$$

Numerical

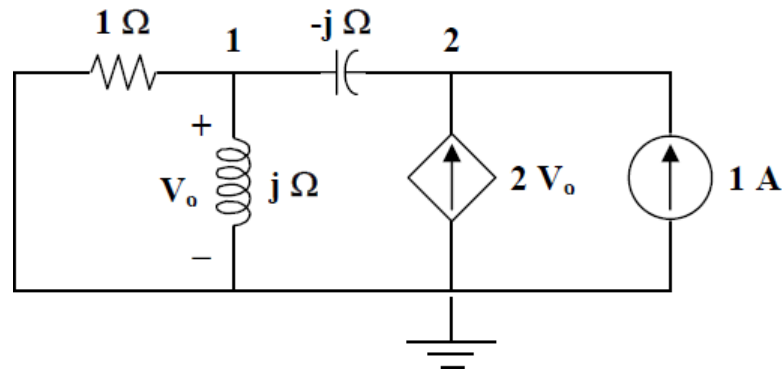
- 2

In the circuit of Fig. find the value of $\mathbf{Z}_L$ that will absorb the maximum power and the value of the maximum power.



- Solution:

To find $\mathbf{Z}_{Th}$, insert a 1-A current source at the load terminals



At node 1,

$$\frac{\mathbf{V}_o}{1} + \frac{\mathbf{V}_o}{j} = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow \mathbf{V}_o = j\mathbf{V}_2 \quad (1)$$

At node 2,

$$1 + 2\mathbf{V}_o = \frac{\mathbf{V}_2 - \mathbf{V}_o}{-j} \longrightarrow 1 = j\mathbf{V}_2 - (2 + j)\mathbf{V}_o \quad (2)$$

Substituting (1) into (2),

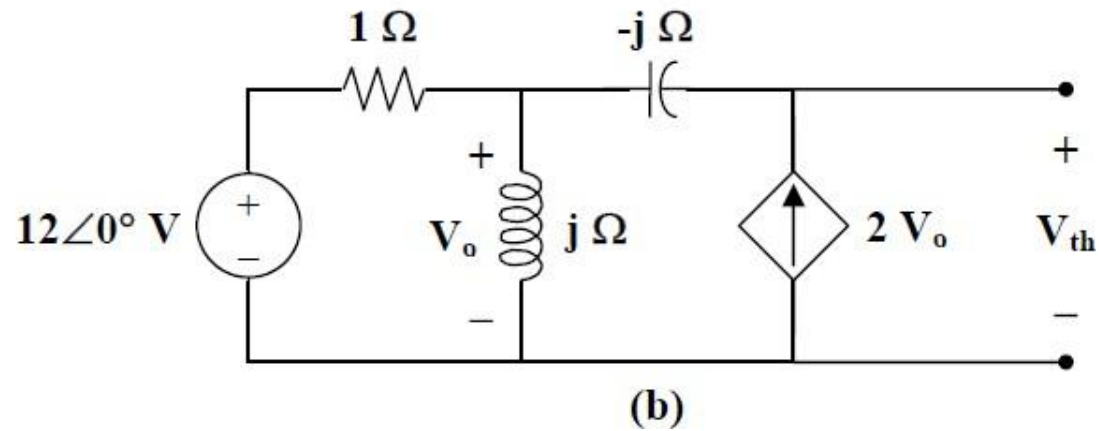
$$1 = j\mathbf{V}_2 - (2 + j)(j)\mathbf{V}_2 = (1 - j)\mathbf{V}_2$$

$$\mathbf{V}_2 = \frac{1}{1 - j}$$

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_2}{1} = \frac{1 + j}{2} = 0.5 + j0.5$$

$$\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = \underline{\underline{0.5 - j0.5 \, \Omega}}$$

We now obtain $\mathbf{V}_{\text{Th}}$ from Fig. (b).



$$2\mathbf{V}_o + \frac{12 - \mathbf{V}_o}{1} = \frac{\mathbf{V}_o}{j}$$

$$\mathbf{V}_o = \frac{-12}{1 + j}$$

$$-\mathbf{V}_o - (-j \times 2\mathbf{V}_o) + \mathbf{V}_{Th} = 0$$

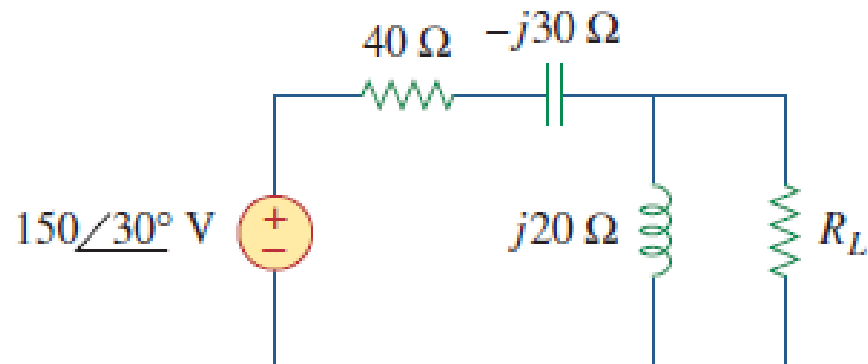
$$\mathbf{V}_{Th} = (1 - j2)\mathbf{V}_o = \frac{(-12)(1 - j2)}{1 + j}$$

$$P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_L} = \frac{\left(\frac{12\sqrt{5}}{\sqrt{2}}\right)^2}{(8)(0.5)} = \underline{\underline{90 \text{ W}}}$$

Find the value of R_L that will absorb the maximum average power.

Calculate the power.

Numerical - 3



Solution:

$$\mathbf{Z}_{\text{Th}} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{j20 + 40 - j30} = 9.412 + j22.35 \, \Omega$$

By voltage division, $\mathbf{V}_{\text{Th}} = \frac{j20}{j20 + 40 - j30} (150 \angle 30^\circ) = 72.76 \angle 134^\circ \text{ V}$

The value of R_L that will absorb the maximum average power is

$$R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{9.412^2 + 22.35^2} = 24.25 \, \Omega$$

The current through the load is

$$\mathbf{I} = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + R_L} = \frac{72.76 \angle 134^\circ}{33.66 + j22.35} = 1.8 \angle 100.42^\circ \text{ A}$$

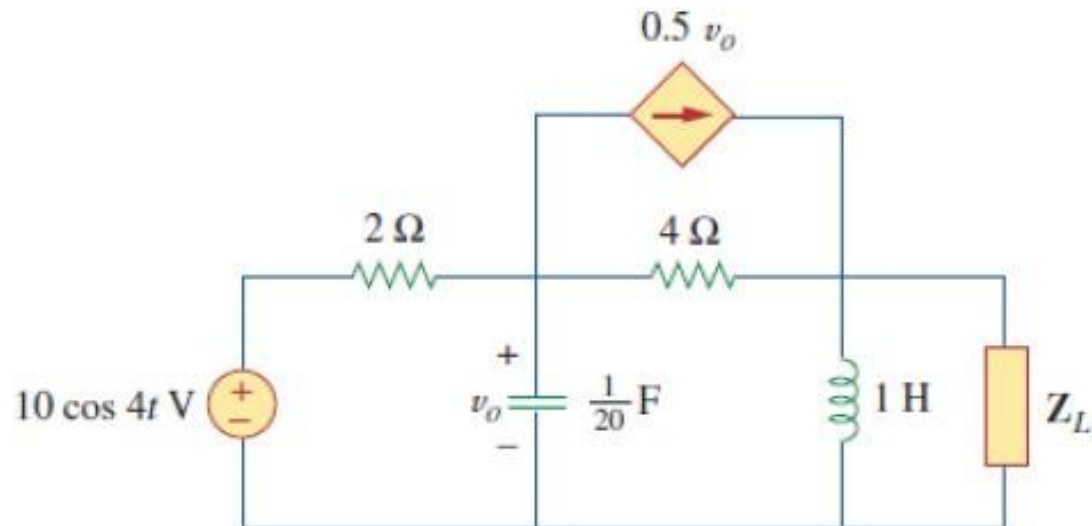
The maximum average power absorbed by R_L is

$$P_{\text{max}} = \frac{1}{2} |\mathbf{I}|^2 R_L = \frac{1}{2} (1.8)^2 (24.25) = 39.29 \text{ W}$$

Numerical

- 4

Determine the maximum average power delivered to the load for the circuit



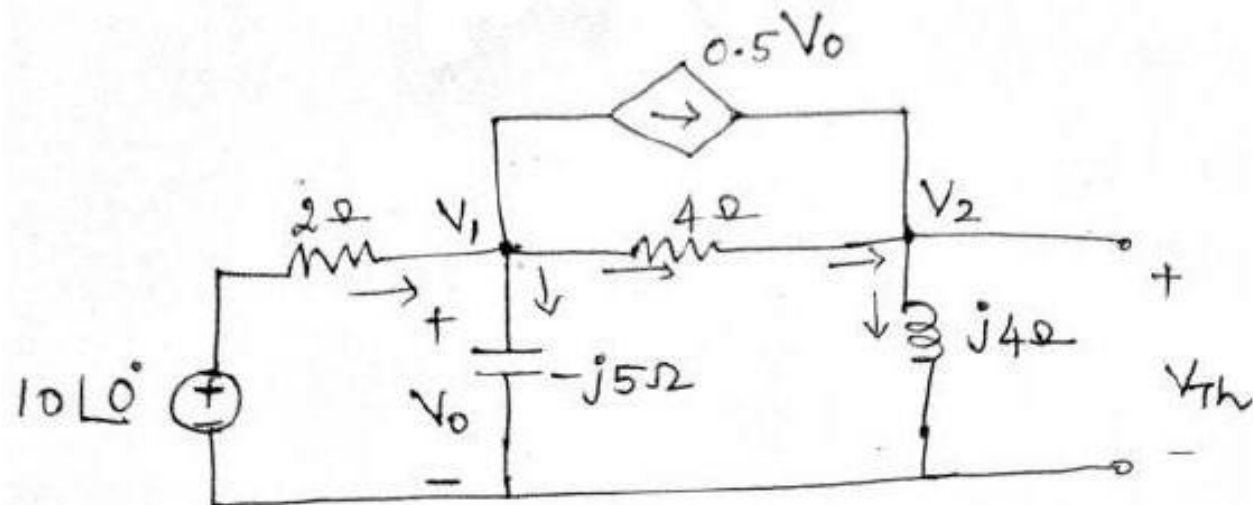
Solution:

$$V_m = 10 ; V = 10 \angle 0^\circ$$

$$\omega = 4 \text{ rad/s}$$

$$1 \text{ H} \Rightarrow j\omega L = j \times 4 \times 1 = j4 \Omega$$

$$\frac{1}{20} \text{ F} \Rightarrow \frac{1}{j\omega C} = \frac{20}{j \times 4} = -j5 \Omega$$



To find V_{th} :-

KCL at node V_1

$$\frac{10 - V_1}{2} = \frac{V_1}{-j5} + 0.5V_0 + \frac{V_1 - V_2}{4} \quad \text{Since } V_0 = V_1$$

$$\frac{10 - V_1}{2} = j0.2V_1 + 0.5V_1 + 0.25(V_1 - V_2)$$

$$5 - 0.5V_1 = j0.2V_1 + 0.5V_1 + 0.25V_1 - 0.25V_2$$

$$1.25V_1 + j0.2V_1 - 0.25V_2 = 5$$

$$\therefore (1.25 + j0.2)V_1 - 0.25V_2 = 5 \rightarrow \textcircled{1}$$

KCL at node V_2 .

$$\frac{(V_1 - V_2)}{4} + 0.5V_1 = \frac{V_2}{j4}$$

$$0.25(V_1 - V_2) + 0.5V_1 = -j0.25V_2$$

$$0.25V_1 - 0.25V_2 + 0.5V_1 + j0.25V_2 = 0$$

$$0.75V_1 + (-0.25 + j0.25)V_2 = 0 \rightarrow \textcircled{2}$$

Solving $\textcircled{1}$ & $\textcircled{2}$

$$\begin{bmatrix} 1.25 + j0.25 & -0.25 \\ 0.75 & -0.25 + j0.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1.25 + j0.2 & -0.25 \\ 0.75 & -0.25 + j0.25 \end{bmatrix}$$

$$= (1.25 + j0.2)(-0.25 + j0.25) + 0.25 \times 0.75$$

$$= -0.3125 + j0.3125 - j0.05 - 0.05 + 0.1875$$

$$\Delta = -0.175 + j0.2625 = 0.3154 \angle 123.7^\circ$$

$$\Delta_2 = \begin{bmatrix} 1.25 + j0.2 & 5 \\ 0.75 & 0 \end{bmatrix}$$

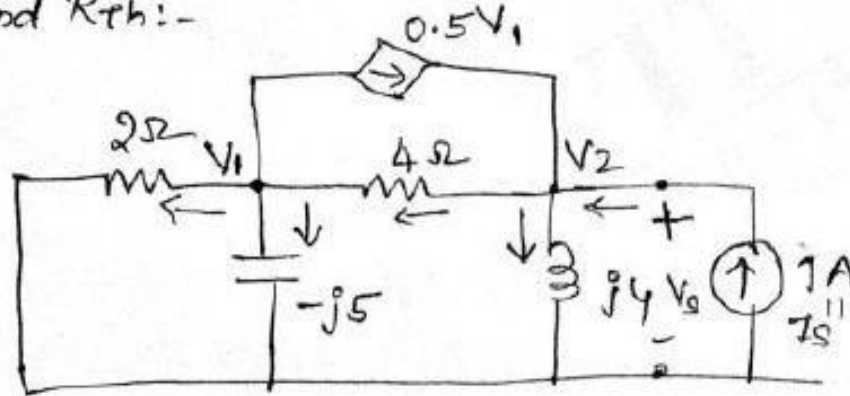
$$\Delta_2 = 0 - 5(0.75) = -3.75 = 3.75 \angle 180^\circ$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{3.75 \angle 180^\circ}{0.3154 \angle 123.7^\circ}$$

$$V_2 = 11.88 \angle 56.3^\circ = (6.59 + j9.88) \text{ V}$$

$$V_{Th} = V_2 = (6.59 + j9.88) \text{ V} = 11.88 \angle 56.3^\circ \text{ V}$$

To find R_{th} :-



KCL at node 1,

$$\frac{V_1}{2} + \frac{V_1}{-j5} + 0.5V_1 = \frac{V_2 - V_1}{4}$$

$$0.5V_1 + j0.2V_1 + 0.5V_1 = 0.25(V_2 - V_1)$$

$$1.25V_1 + j0.2V_1 - 0.25V_2 = 0$$

$$(1.25 + j0.2)V_1 - 0.25V_2 = 0 \quad \text{--- (3)}$$

KCL at node 2,

$$1 + 0.5V_1 = \frac{V_2 - V_1}{4} + \frac{V_2}{j4}$$

$$1 + 0.5V_1 = 0.25(V_2 - V_1) - j0.25V_2$$

$$1 = 0.25V_2 - 0.25V_1 - 0.5V_1 - j0.25V_2$$

$$-0.75V_1 + (0.25 - j0.25)V_2 = 1 \rightarrow \textcircled{3}$$

Solving $\textcircled{3}$ & $\textcircled{4}$

$$\textcircled{4} \Rightarrow 0.75V_1 + (-0.25 + j0.25)V_2 = -1 \rightarrow \textcircled{4}$$

$$\Delta = 0.3154 \angle 123.7^\circ$$

$$\Delta_2 = \begin{bmatrix} 1.25 + j0.25 & 0 \\ 0.75 & -1 \end{bmatrix}$$

$$= -1.25 - j0.25 = 1.27 \angle -170.9^\circ$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{1.27 \angle -170.9}{0.3154 \angle 123.7}$$

$$V_2 = 4.026 \angle -294.6 = 4.026 \angle 65.4^\circ$$

$$V_2 = V_g$$

$$Z_{Th} = \frac{V_g}{I_g} = \frac{4.026 \angle 65.4^\circ}{1} = 4.026 \angle 65.4^\circ$$

$$\therefore Z_{Th} = 4.026 \angle 65.4^\circ = 1.675 + j3.66 \, \Omega$$

$$\therefore P_{max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{(11.88)^2}{8(1.675)} = \frac{141.13}{13.4}$$

$$\boxed{P_{max} = 10.53 \text{ W}}$$

Quick comparison

Quantity	Formula	Meaning	Unit
Instantaneous power	$p(t) = v(t)i(t)$	Power at a particular instant	W
Average power	$P = \frac{1}{T} \int_0^T p(t) dt$	Average/actual power consumed	W
Reactive power	$Q = VI \sin \phi$	Energy exchanged with L/C	VAR
Apparent power	(	S	=VI)
Complex power	$S = P + jQ$	Combines real and reactive power	VA

$$\text{Power Factor} = \frac{\text{Real Power}}{\text{Apparent Power}}$$

$$\text{Power Factor} = \cos \phi$$

Thank You